PROJECT REPORT · AI IN VIRTUAL CARE

Who gets left out of telehealth, and where AI fits

Rural and urban Medicare telehealth use, 2020 to 2025: what the CMS claims show, what explains the gap, how AI is being used to close it, and a forecast model with a fairness audit.

Neha · October 2026 · Data: CMS Medicare Telehealth Trends, fee-for-service Part B claims through 2025-Q4 (2025-Q4 preliminary)

Summary

AI plays three roles in this project: a working tool for the analysis, a set of interventions we study, and a forecasting model we built and audited.

- **AI as analysis tool (done).** Claude cleaned the CMS Medicare Telehealth Trends file, built the rural-urban analysis in R, and tested outside explanations.
- **AI as subject (researched).** Voice assistants, automated interpreting, ambient scribes and AI-enabled remote monitoring are being used to widen access. The evidence is mostly small pilots.
- **AI as something to build (built).** A forecasting model with a fairness audit. A seasonal baseline beat the machine-learning models; forecasts are equally accurate for rural and urban beneficiaries but less accurate for those 85 and over and in low-broadband states.

10.6 → 7.3
Rural-urban gap in points, 2020 to 2024

3.2 + 4.1
2024 gap: within states + where rural beneficiaries live

38 of 49
States where the gap narrowed, 2021 to 2025

0.07
Correlation of 65+ broadband with the gap (0.59 with overall use)

The data question behind all three: rural Medicare beneficiaries use telehealth less than urban ones (a 10.6-point gap in 2020, 7.3 in 2024). About 3.2 points of the 2024 gap is within states, and about 4.1 points comes from rural beneficiaries living in states where everyone uses less. Broadband tracks how much telehealth a state uses, but not the size of its gap.

AI across the project

Each finding about who is left out maps to an AI approach, and Claude did the analysis itself; the forecasting and fairness-audit stages are built, with results below.

PROJECT STAGE	WHAT WE FOUND OR DID	ROLE OF AI	STATUS
Who is left out	Rural use trails urban by 7.3 points (2024); over half of that comes from where rural beneficiaries live	Ambient scribes could free rural clinician time; the evidence so far is facility-reported	Researched
Why the gap exists	Broadband tracks a state's overall use, not the gap; nothing measured explains the gap inside states	Voice AI over ordinary phone calls for people without reliable video; usability pilots only	Researched
Language and age barriers	Patients 65 and older and Spanish speakers were less likely to have video visits (California study)	Automated interpreting (114-patient trial) and accent-robust speech recognition (small pilot)	Researched
Care at a distance	Remote monitoring trial found worse results at 65 and older	AI-enabled monitoring, tested by age group before scaling	Researched
Forecasting	Quarterly rural and urban rates by state through 2025-Q3	A forecasting model of state and group telehealth use; a seasonal baseline beat the random forest	Built
Fairness audit	Gaps differ by state size, rurality and broadband	Checks of whether forecasts err more for rural, older or low-broadband groups: rural fine, ages 85 and over and low-broadband states weaker	Built
Doing the analysis	36,120 CMS rows cleaned, state and national gaps analyzed, county data merged	Claude cleaned, analyzed in R and merged the outside data; a person reviews numbers, sources and caveats	Done

What the data shows

The rural-urban gap has narrowed since 2020, but more than half of the 2024 gap comes from where rural beneficiaries live, and nothing we could measure

explains the part inside states.

> Measure: share of telehealth-eligible Medicare fee-for-service beneficiaries with at least one telehealth service (audio-only included). Gap = urban minus rural, in percentage points. Rural and urban come from the beneficiary's ZIP code (RUCA).

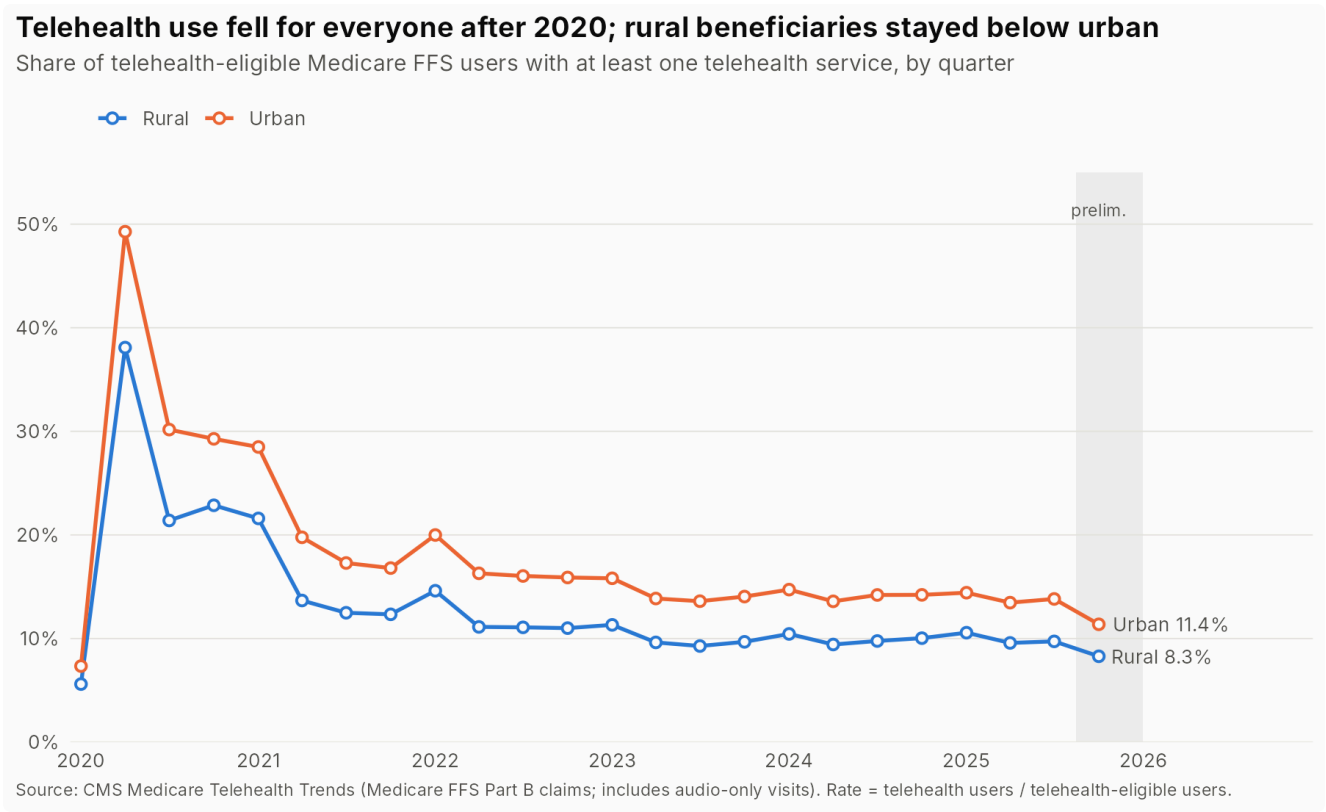


Figure 1. Use fell for everyone after 2020, and rural beneficiaries stayed below urban.

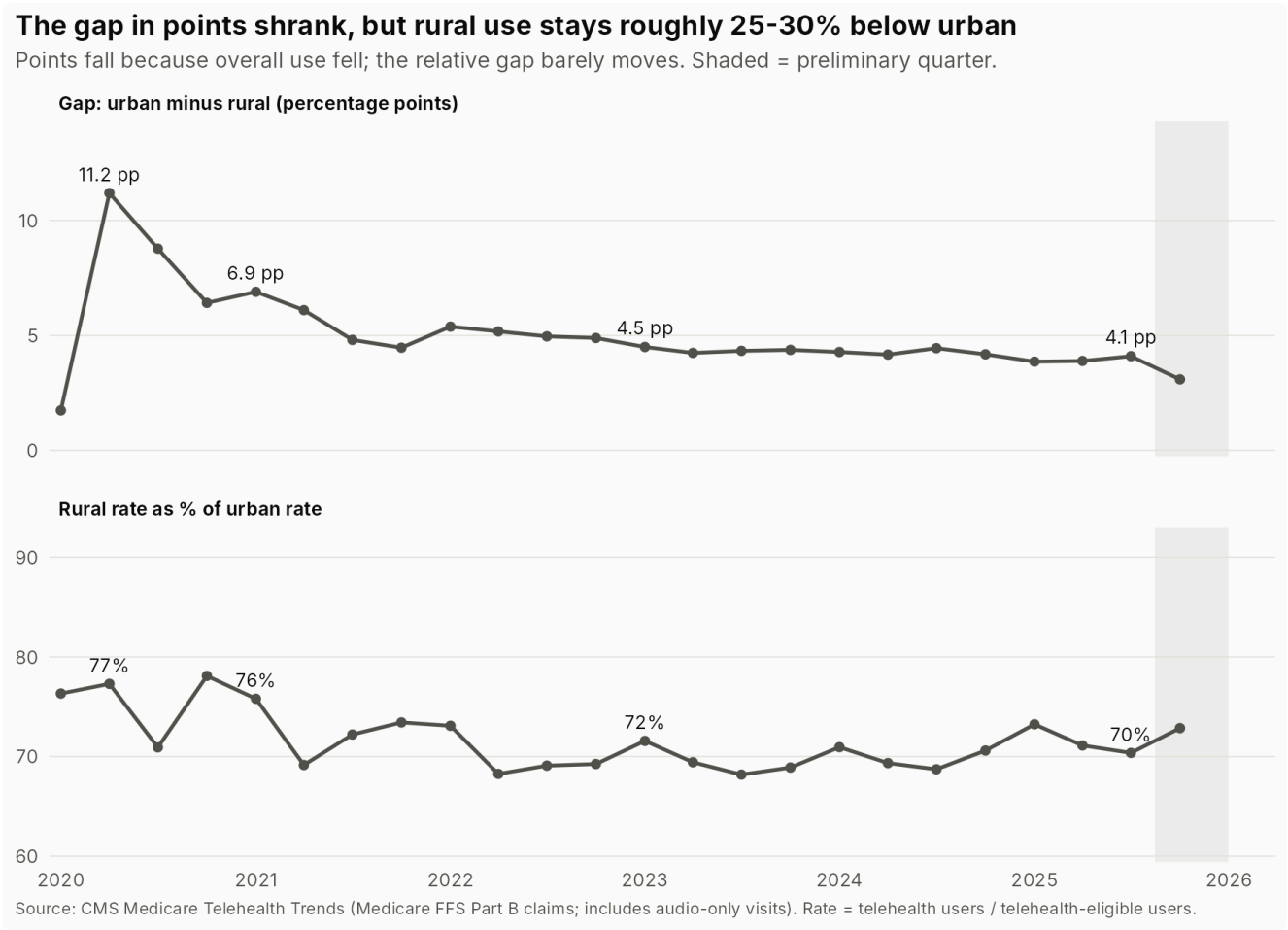


Figure 2. The gap in points shrank, but rural use stays roughly 25–30% below urban. Points fall because overall use fell; the relative gap barely moves.

The national gap fell from 11.2 points in 2020-Q2 to 4.1 points in 2025-Q3, a decline of 0.43 points a year (95% interval −0.61 to −0.26) over 2021-Q1 to 2025-Q3.

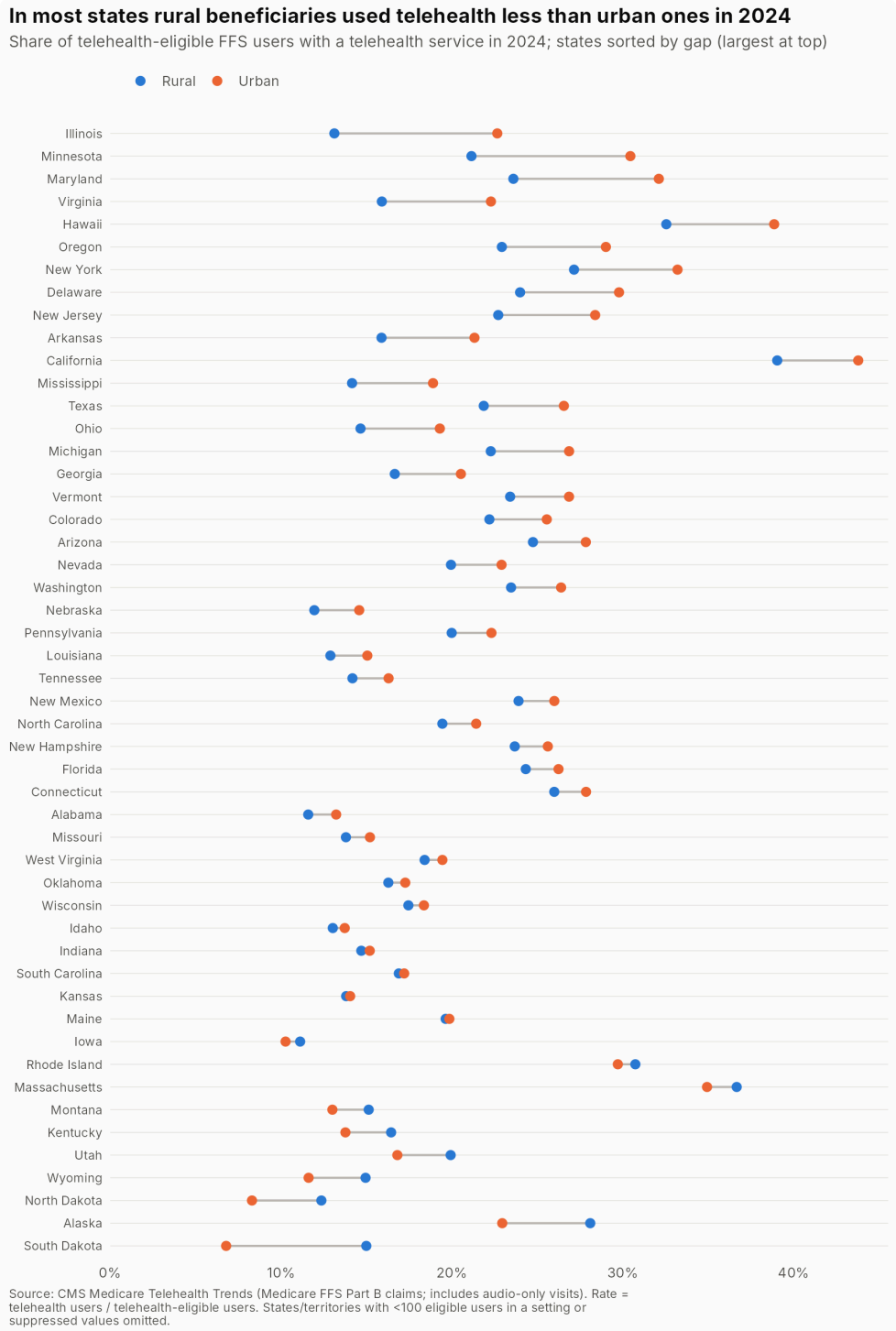


Figure 3. In most states rural beneficiaries used telehealth less than urban ones in 2024. Urban was higher in 40 states; the median gap was 2.1 points, ranging from −8.2 to 9.5.

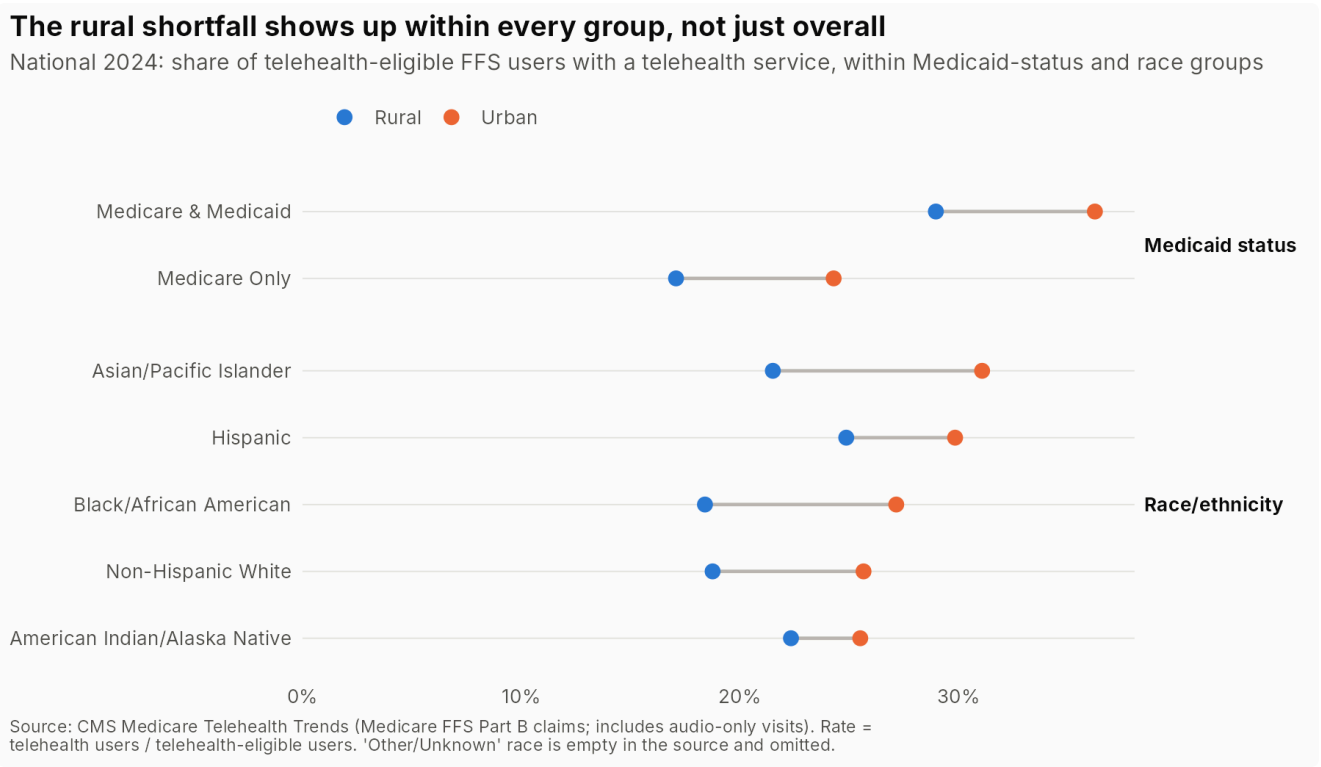


Figure 4. The rural shortfall shows up within every group, not just overall. After adjusting for Medicaid status and race, the national gap is about 7.0–7.2 points versus 7.3 unadjusted.

State trends, 2021 to 2025

Per-state trends use a linear fit with quarter-of-year effects and Newey-West intervals (lag 3). 49 states had enough usable quarters; the District of Columbia and Rhode Island were excluded.

- **Gap shrinking in most states.** The gap narrowed in 38 of 49 states (6 no clear trend, 5 widened), and rural use is catching up relative to urban in 30 of 49 (8 falling behind, 11 no clear trend).
- **Urban advantage is widespread.** Urban was higher in nearly every quarter in 36 states, rural higher in 5, and 8 crossed back and forth.

STATE	AVERAGE GAP, 2021–25 (PTS)	GAP IN 2021 (PTS)	GAP IN 2025 (PTS)	CHANGE PER YEAR (PTS)	TREND
Minnesota	7.1	9.4	5.9	-0.89 (-1.33 to -0.45)	Narrowing
Maryland	6.4	8.5	5.1	-1.02 (-1.34 to -0.70)	Narrowing
Illinois	5.7	5.9	5.4	-0.15 (-0.28 to -0.02)	Narrowing
Oregon	4.5	6.1	3.6	-0.70 (-0.92 to -0.48)	Narrowing
Michigan	4.2	5.7	2.6	-0.83 (-0.92 to -0.75)	Narrowing
Delaware	4.2	5.9	2.9	-0.58 (-1.03 to -0.12)	Narrowing
Virginia	3.9	5.1	3.4	-0.45 (-0.72 to -0.17)	Narrowing
Hawaii	3.8	6.7	1.3	-1.42 (-1.74 to -1.10)	Narrowing

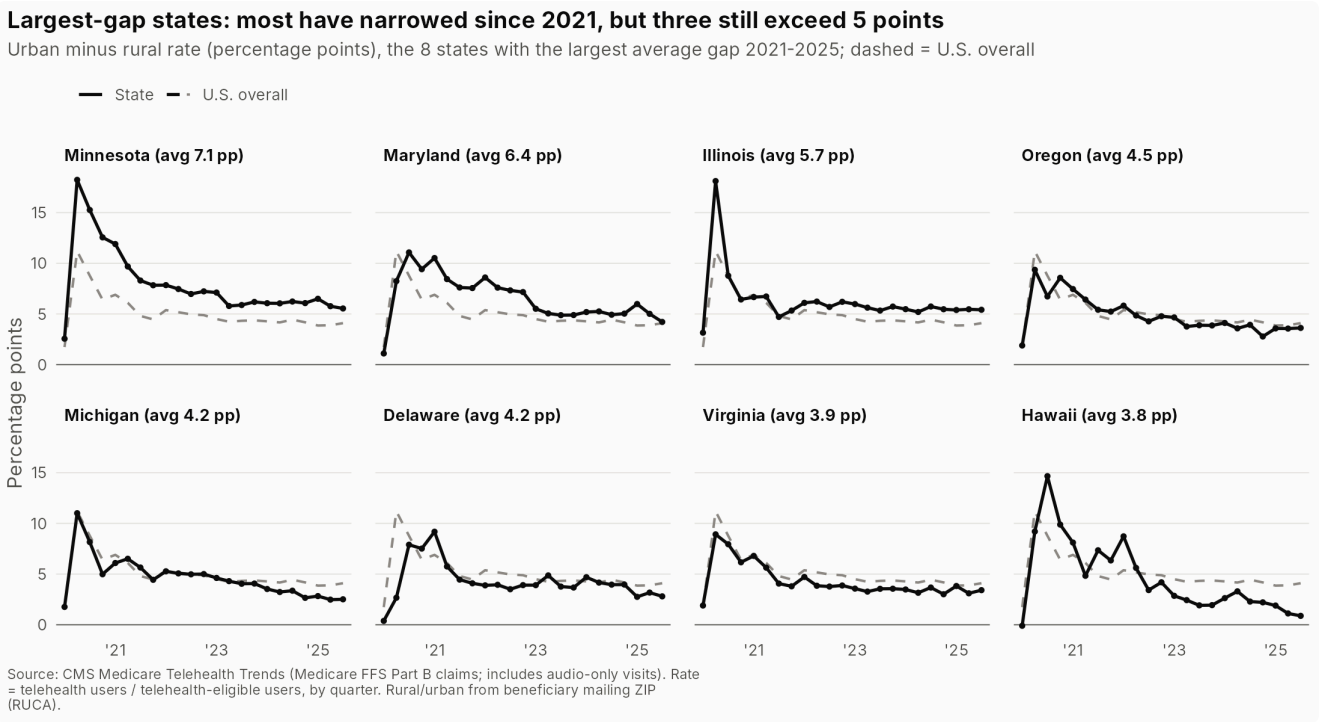


Figure 5. Largest-gap states: most have narrowed since 2021, but three (Minnesota, Maryland, Illinois) still exceed 5 points in 2025.

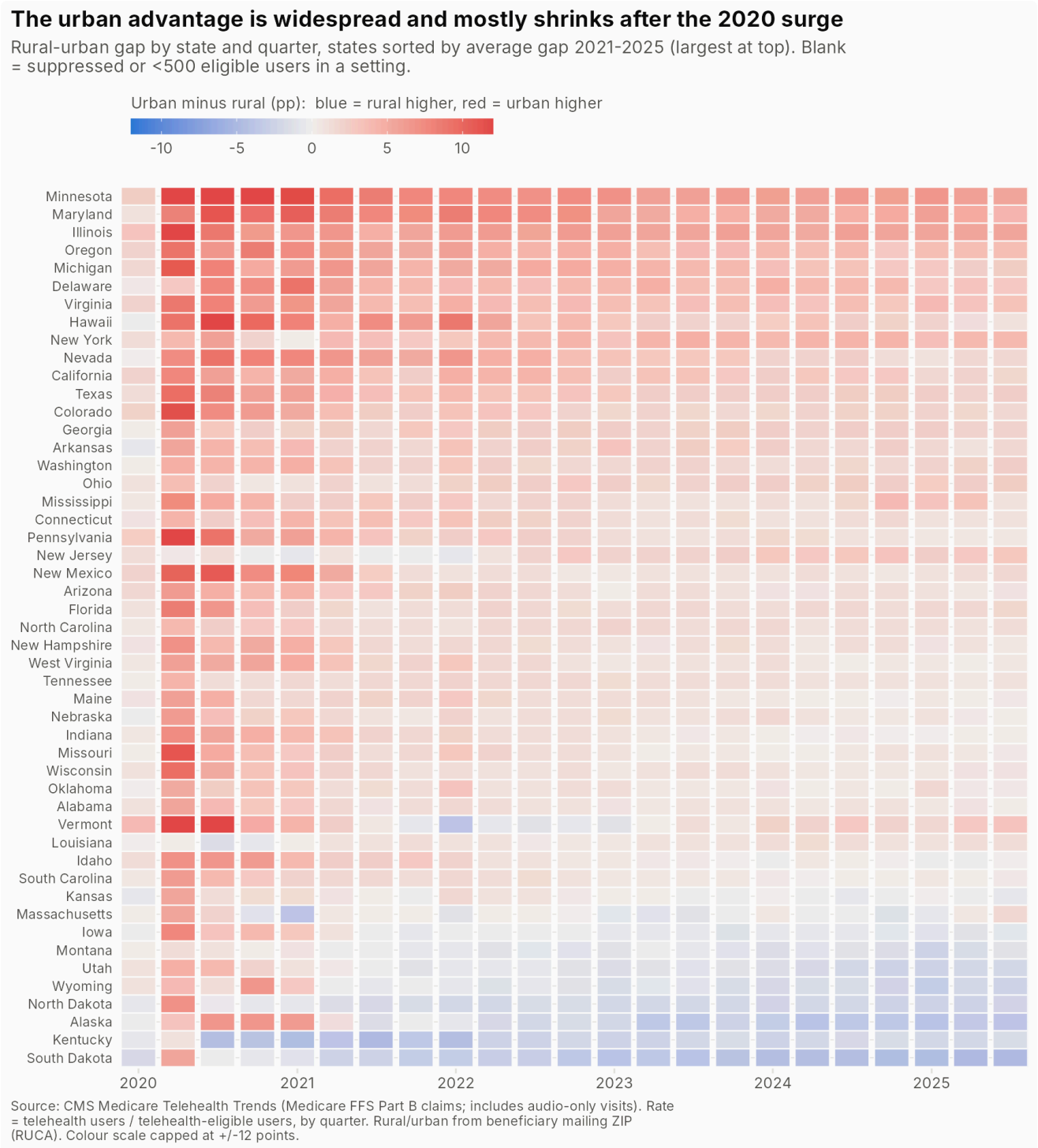


Figure 6. The urban advantage is widespread and mostly shrinks after the 2020 surge. Blue cells mark states where rural use is higher.



Figure 7. Gaps shrink in points almost everywhere, but relative catch-up is mixed. With 49 states, a few will look “clear” by chance.

Why state trends and the national ratio disagree

Rural use is about 70–73% of urban use nationally, with no clear trend, even though rural beneficiaries are catching up inside most states.

- **National ratio flat.** The trend is −0.003 a year (interval −0.010 to 0.004).
- **Inside states, rural catches up** by 3.8 log points. **Urban rates fell more in states with many rural beneficiaries**, which takes off 5.9. The two roughly cancel, which is why the national ratio does not move.

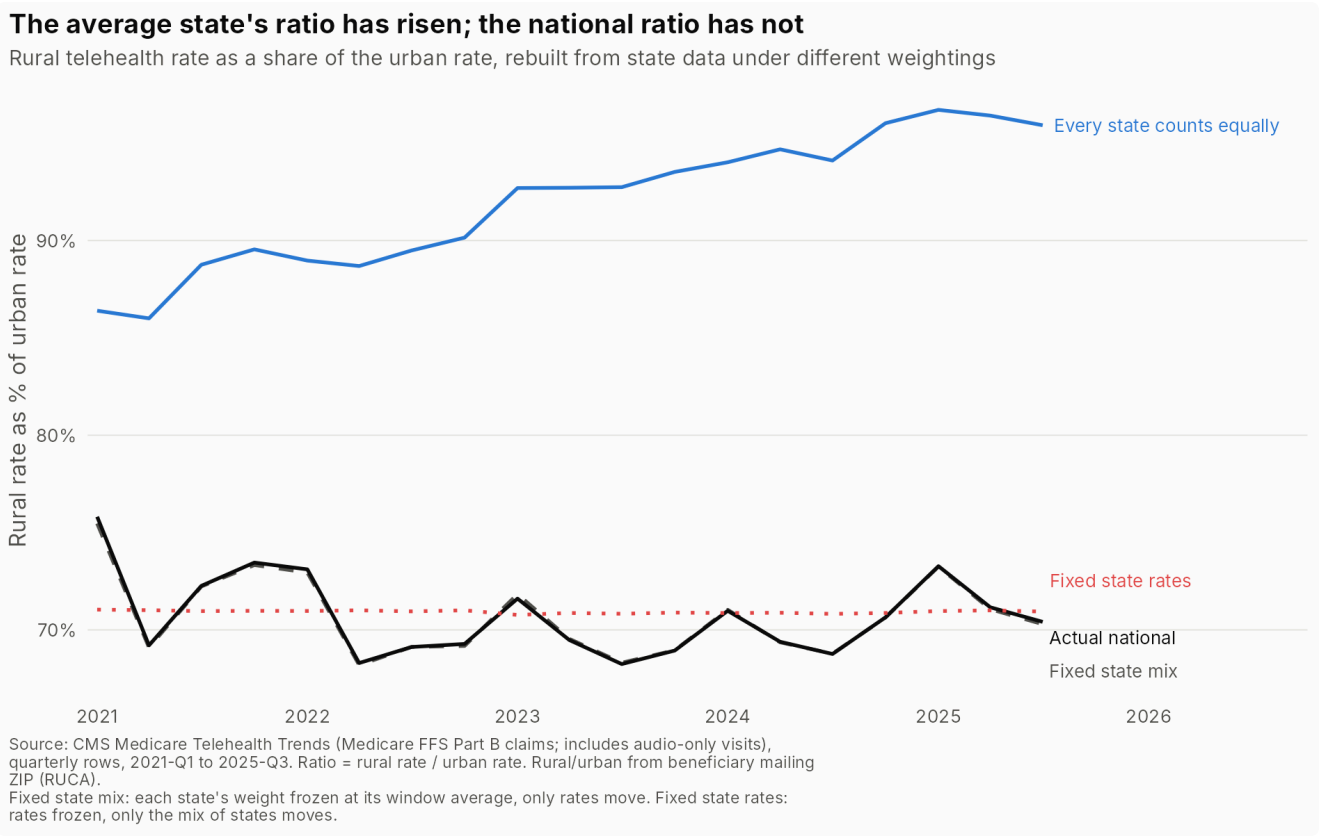


Figure 8. The average state’s ratio has risen; the national ratio has not.

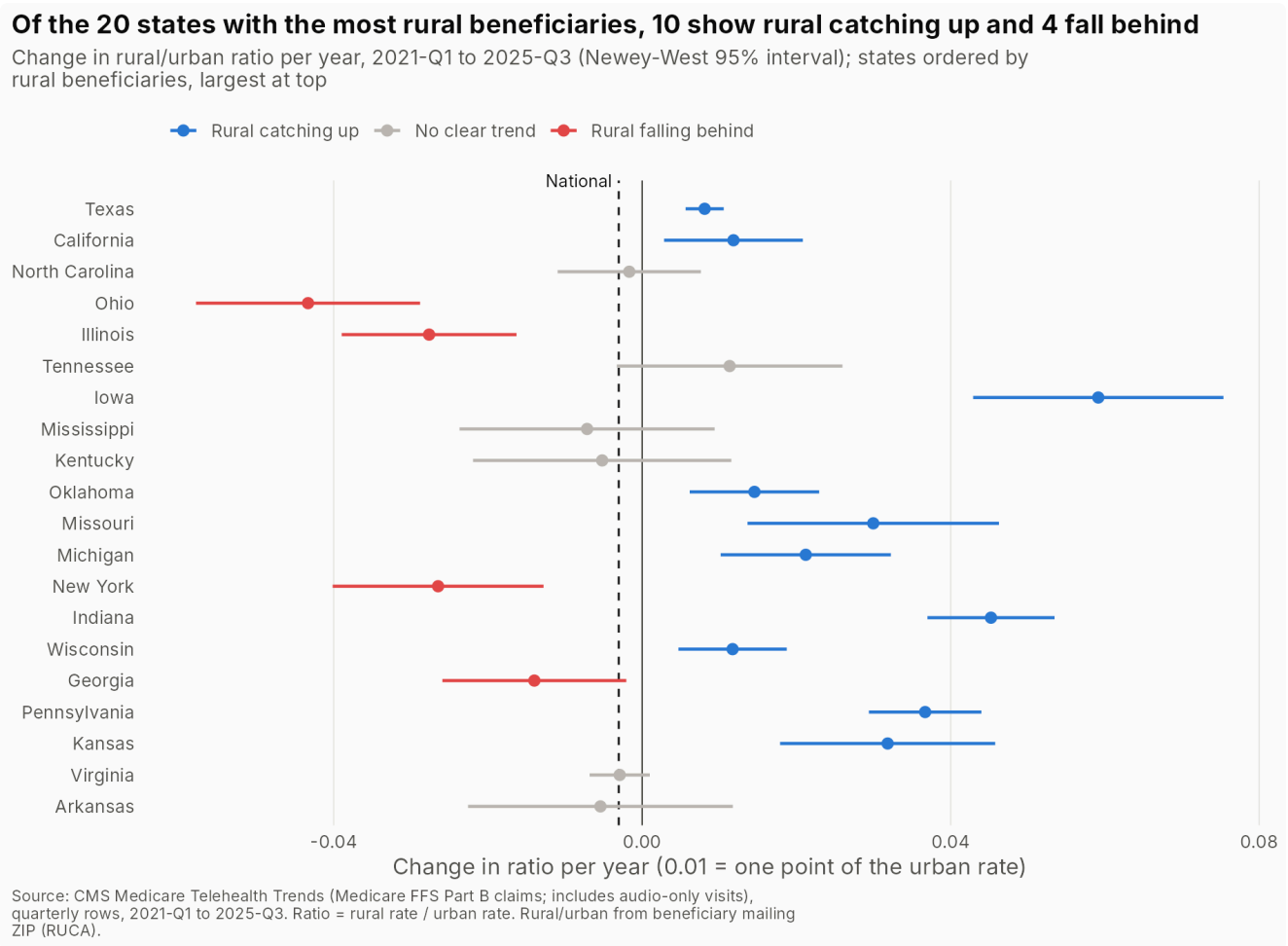


Figure 9. Ratio change per year in the 20 states with the most rural beneficiaries, with Newey-West intervals.

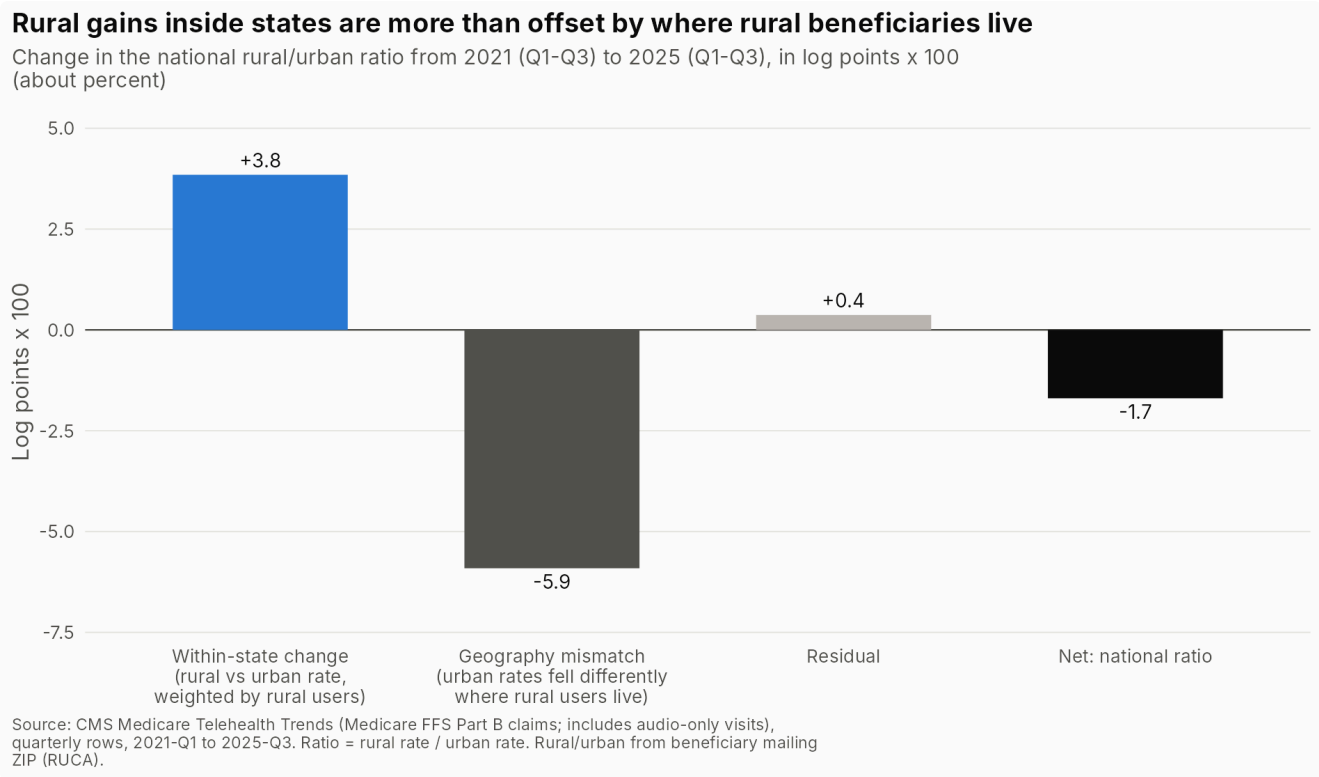


Figure 10. Rural gains inside states are more than offset by where rural beneficiaries live.

What explains the gap

The 2024 gap splits into a within-state part and a “place” part. Outside data on broadband, physician supply and Medicare Advantage tracks how much telehealth a state uses, not the size of its gap.

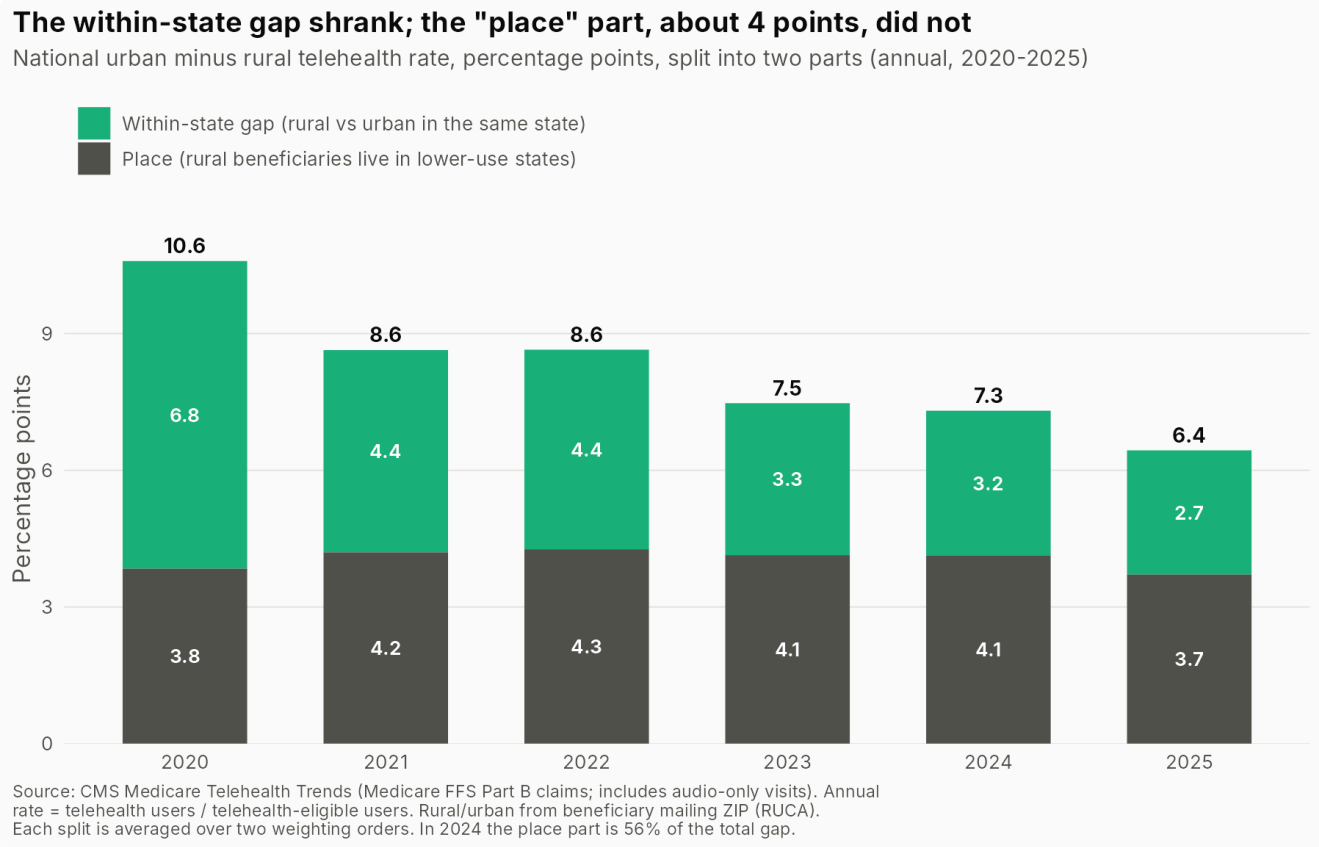


Figure 11. The within-state gap shrank; the “place” part, about 4 points, did not. In 2024 the place part is 56% of the 7.3-point gap.

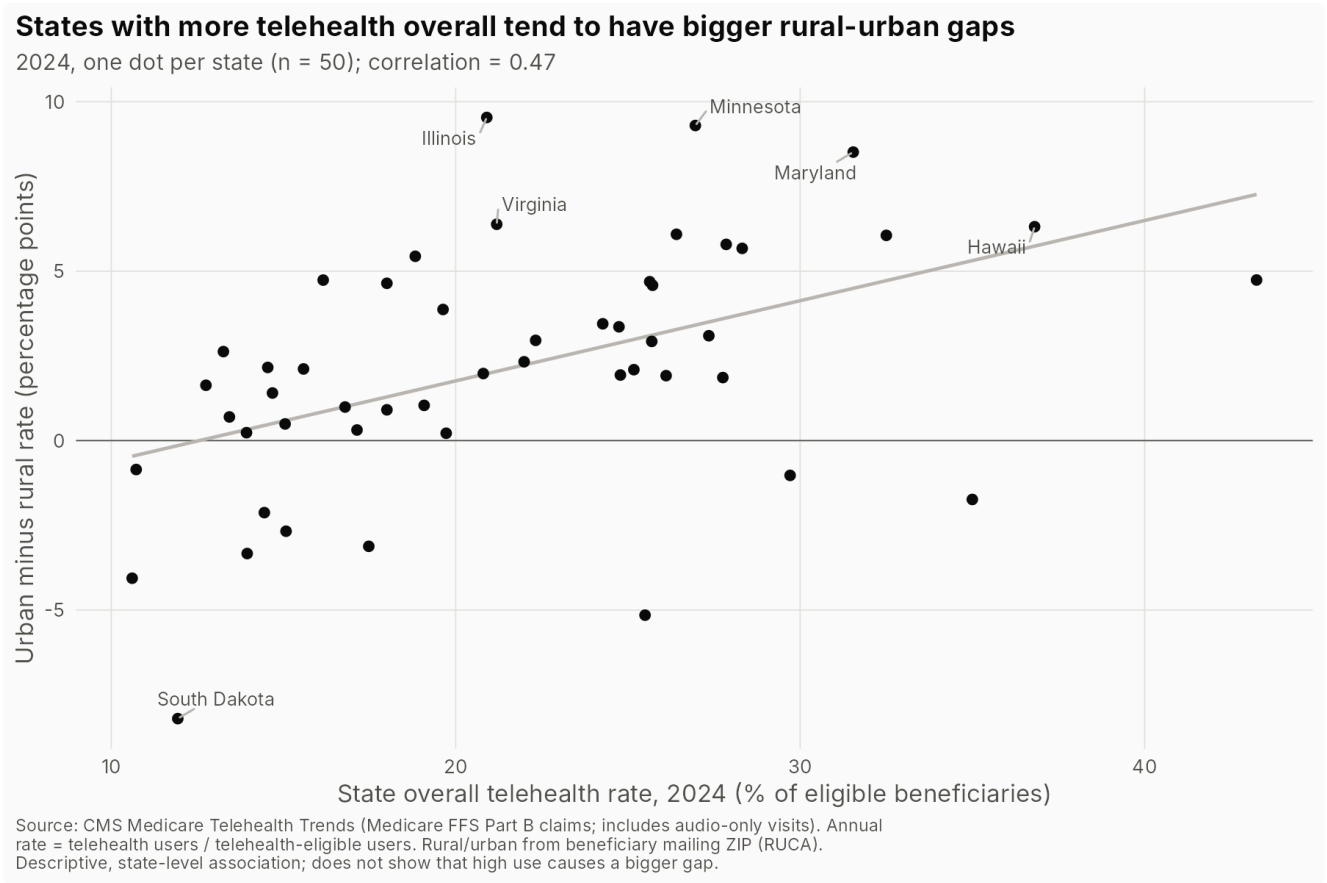


Figure 12. States with more telehealth overall tend to have bigger rural-urban gaps (correlation 0.47, 50 states). Descriptive only.

Outside data

- **Broadband tracks level, not gap.** Across states, 65+ broadband correlates with rural use (r = 0.59), urban use (0.42) and overall use (0.59), but not with the gap (0.07). Primary-care supply and Medicare Advantage share don't line up with the gap either (0.06 and 0.09).
- **Rural counties do trail on the inputs.** 75.7% of 65+ households have a computer and broadband versus 84.0% in urban counties; 5.1 primary-care physicians per 10,000 people versus 7.9.

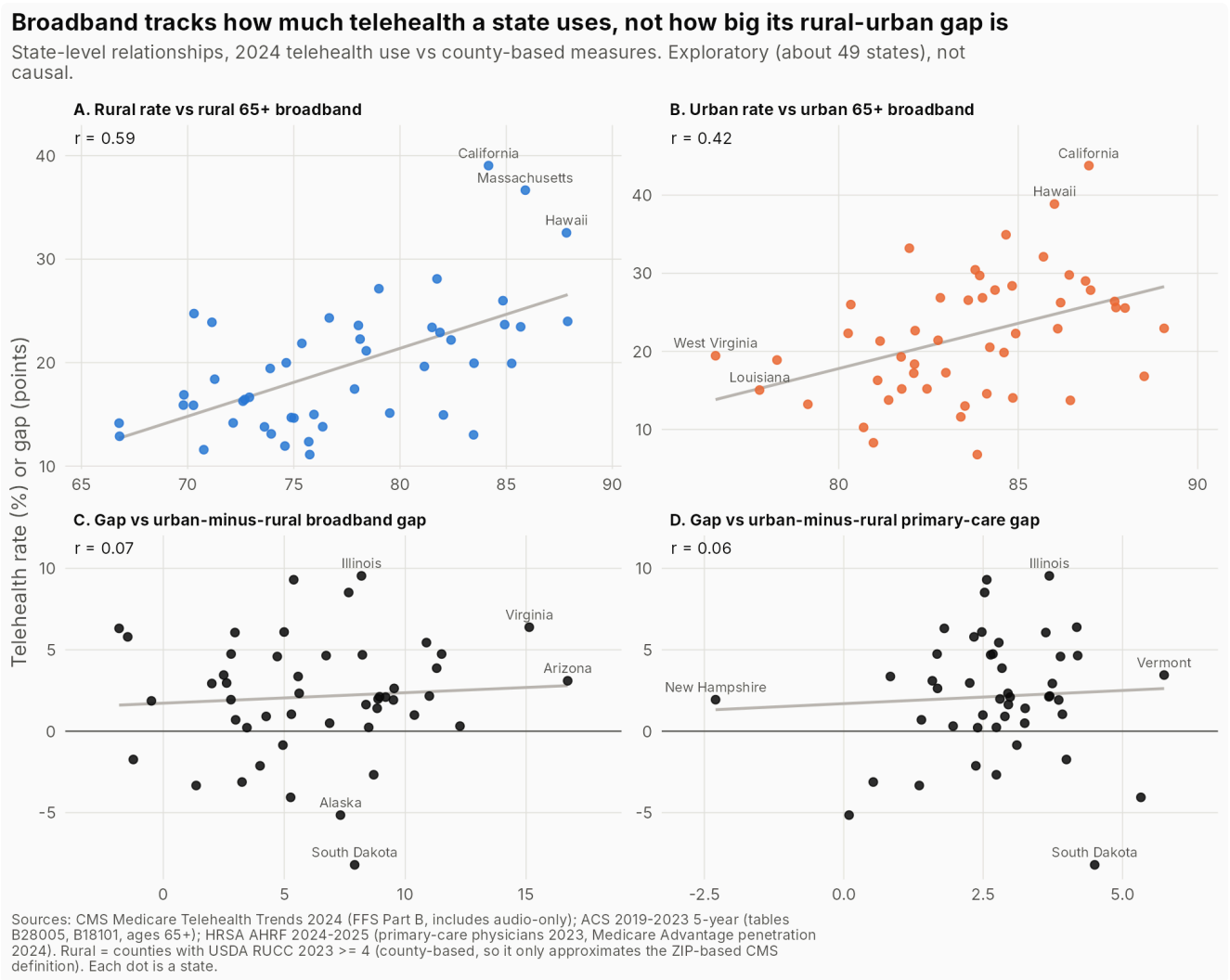


Figure 13. Broadband tracks how much telehealth a state uses, not how big its rural-urban gap is. Exploratory, about 49 states, not causal.

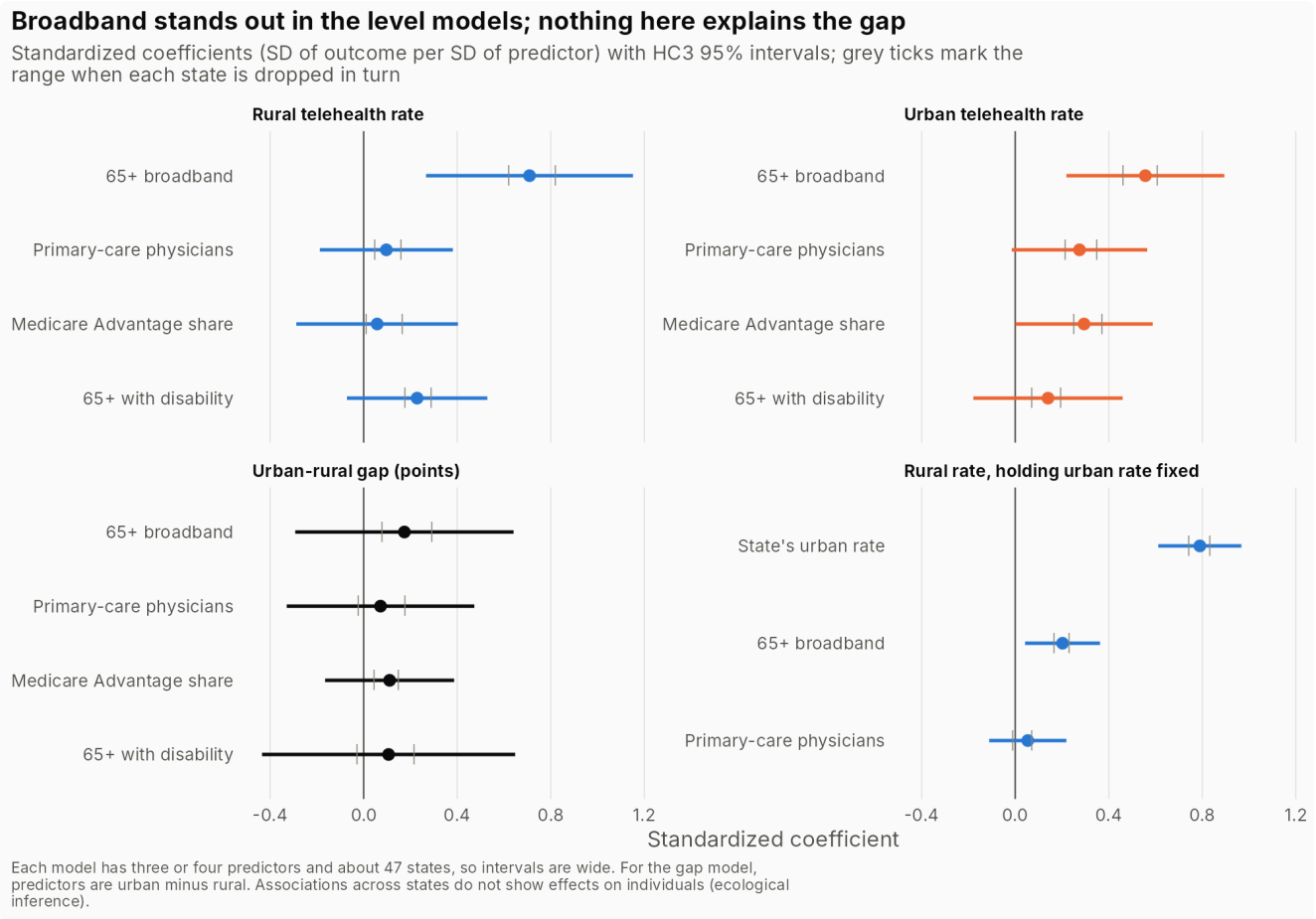


Figure 14. Broadband stands out in the level models; nothing here explains the gap. Standardized coefficients with HC3 95% intervals.

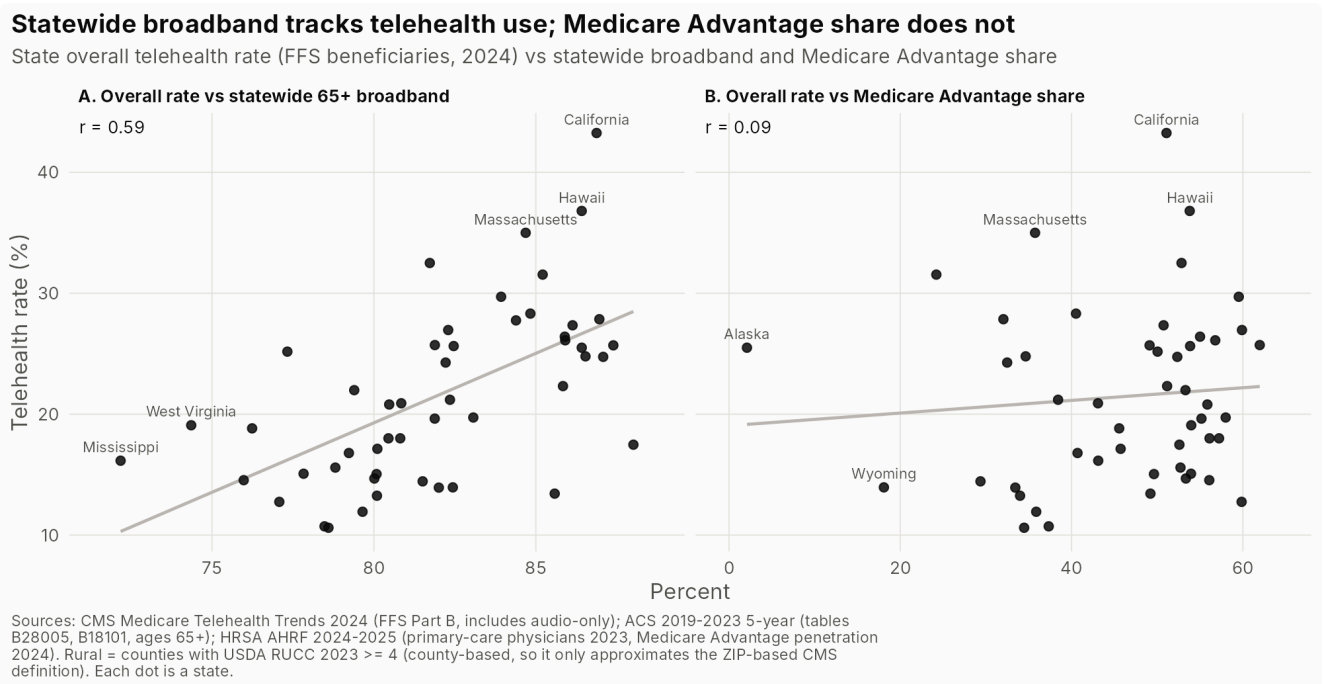


Figure 15. Statewide broadband tracks telehealth use; Medicare Advantage share does not.

> These are comparisons across about 49 states, so they show association only. Rural is defined by county (USDA codes) in the outside data and by ZIP code in the CMS file.

AI as a subject

AI tools are being used to widen telehealth access, but the evidence that they close gaps like ours is thin and early, and some of them risk widening the gap.

How AI is being used to close the gap

AI is being tried on four parts of the problem, but none of the evidence found shows it narrowing the rural-urban gap.

APPROACH	WHAT IT TARGETS	BEST EVIDENCE	STRENGTH OF EVIDENCE
Voice AI over phone calls	People who can't or won't use video	Home-hospital voice assistant tested with 18 clinicians and 19 patients; Talk2Care for older adults	Usability only, no clinical outcomes
Automated interpreting	Language barriers	AHRQ-funded trial: 114 Spanish-speaking patients; accuracy comparable to interpreters reported in the literature, weaker on figurative language	Randomized, some aims unfinished
Ambient AI scribes	Rural clinician shortage and burnout	Rural facilities report documentation falling from 20 to 12 minutes per patient; Becker's cites a 263-clinician study where burnout fell from 51.9% to 38.8%	Self-reported; the study not checked at source
AI-enabled remote monitoring	Chronic care at a distance	A randomized trial of 1,286 discharged adults found no gain in days at home, and fewer home days and more readmissions at 65+	Remote monitoring in general, not AI-specific

Policy matters alongside the technology. Medicare's telehealth flexibilities, including home visits and audio-only coverage, were extended through December 31, 2027.

Where AI can make the gap worse

- **Accent and language errors.** In a small pilot (20 participants reading scripts), speech recognition made more errors for non-native English speakers; a GPT-4o correction step cut them by 4.5 to 6.9 points. The rural AI toolkit also warns of weaker scribe performance for other languages and regional accents, plus missed details and wrong speaker labels.
- **Weak equity evidence.** A review of 50 studies found equity benefits "not automatic" and equity outcomes poorly reported.
- **Unequal access to video.** In a California study of 9,359 rural patients (not about AI), patients 65 and older, Latino and Spanish-speaking patients, and those with public insurance were less likely to have video visits. Tools that assume video leave them out.
- **Monitoring that may not help older adults.** The remote-monitoring trial found worse results at 65 and older, a reminder to test by age group before scaling.

What it means for this project

The studies measure usability and clinician time, not whether patients end up with better access, so our own results are the best guide to where AI could matter.

- **Connectivity.** Broadband tracked overall use but not the gap, so tools that get people online or onto video may lift use without narrowing the gap (our inference). Voice AI over ordinary phone calls reaches the people who stay off video.
- **Clinic capacity.** Scribes are the most plausible rural lever, but no study found links them to rural patients' telehealth use.
- **Who to test on.** The California study and the monitoring trial both point to age and language, so any tool should be checked by age, language and rural status. That is the job of the fairness audit below.
- **Outcomes to measure.** Equity claims need group-level results, such as error rates, completed visits and follow-up by group, not satisfaction averages.

The forecasting model and fairness audit

A seasonal baseline forecasts telehealth use as accurately as the machine-learning models, and no model's errors fall harder on rural beneficiaries. They do fall harder on beneficiaries 85 and over and in low-broadband states.

What was built

- **Data.** Quarterly telehealth rates (users divided by eligible beneficiaries), 2021-Q1 to 2025-Q3, for two panels: state by rural or urban (100 cells) and state by age group (204 cells). Cells under 500 eligible users in any quarter were dropped. Predictors were earlier quarters, season, cell size and the county broadband, physician and Medicare Advantage measures.
- **Models.** Three simple baselines (seasonal naive, seasonal naive plus the latest change, last value) and two learners trained to predict change from the same quarter a year earlier: a lasso and a random forest. The random forest was named the headline model in advance.
- **Holdout.** Forecasts made at the end of 2024-Q3, trained only on data up to then, for the next four quarters. A second run from the end of 2023-Q3 checks that the result holds.
- **Audit.** For each group, forecast error as a share of actual use, compared with all other cells in the same panel (the error ratio), plus bias (over or under-prediction). Intervals come from resampling states 2,000 times.
- **Rules, fixed before any results were seen.** (R1) The headline model must be no worse than the best baseline, or the baseline is preferred. (R2) No group's error ratio may exceed 1.25.

(R3) No group’s bias may exceed 10%. A group fails if its estimate breaks a rule, is uncertain if the estimate passes but its interval crosses the limit, and passes otherwise.

Results

The random forest failed R1 in both panels. Its error was 7.3% and 8.1% of actual use, against 4.7% for the seasonal baseline, so the baseline is the preferred model. From the earlier origin, last value was best and every learner was worse than it, so no learner beat the best simple baseline in either run.

MODEL (ERROR AS % OF ACTUAL USE, 4 QUARTERS AHEAD)	STATE BY RURAL/URBAN	STATE BY AGE	BIAS, RURAL/URBAN	BIAS, AGE
Seasonal naive	4.7%	4.7%	+0.5%	−0.7%
Lasso	4.9%	5.0%	−1.7%	−2.0%
Last value	5.4%	5.3%	−0.4%	−1.5%
Seasonal naive plus latest change	6.6%	6.2%	+4.0%	+3.0%
Random forest (headline)	**7.3%**	**8.1%**	**−6.5%**	**−7.5%**

An error ratio of 1 means the group is forecast as accurately as everyone else; above 1.25 breaks the rule.

GROUP	CELLS	SEASONAL BASELINE: RATIO (95% INTERVAL)	STATUS	RANDOM FOREST: RATIO (95% INTERVAL)	STATUS
Rural	49	0.89 (0.76 to 1.07)	Pass	0.96 (0.83 to 1.11)	Pass
Urban	51	1.13 (0.93 to 1.33)	Uncertain	1.04 (0.90 to 1.20)	Pass
Ages 65–74	51	0.91 (0.77 to 1.04)	Pass	1.07 (0.99 to 1.14)	Pass
Ages 75–84	51	1.15 (1.00 to 1.31)	Uncertain	0.94 (0.83 to 1.06)	Pass
Ages 85 and over	51	1.56 (1.35 to 1.82)	Fail	1.17 (0.99 to 1.35)	Uncertain
Low-broadband states, rural/urban panel	34	1.70 (0.91 to 3.19)	Fail	1.36 (0.81 to 2.22)	Fail
Low-broadband states, age panel	68	1.54 (0.94 to 2.66)	Fail	1.20 (0.86 to 1.71)	Uncertain

- **Rural is served as well as urban.** Rural cells are forecast at least as accurately as urban ones by both models, and rural error ratios are below 1.
- **Ages 85 and over are the clearest weak spot.** The baseline's error for them is 56% above everyone else's, and even the low end of its interval (1.35) is above the 1.25 limit. The random forest narrows the gap (1.17) but its interval still crosses the limit.
- **Low-broadband states look worse, but the intervals are wide.** The estimates break the 1.25 rule in 3 of 4 cases, yet every interval also includes 1, so this is a flag to follow up rather than a finding.
- **Bias.** The baseline is within 3.5% in every group shown. The random forest under-predicts by 6–9% everywhere, inside the 10% limit; from the earlier origin the learners under-predicted by about 11% in all groups alike, which is a model-wide problem rather than a group-specific one.

> Cautions: small cells are noisier, so some of the gap for the 85-and-over group and for low-broadband states may be small-number noise; about 20 group comparisons were run,

> so one or two could look bad by chance; there is a single holdout year, a period when rates were falling; and the model describes forecast accuracy, not why groups differ. The CMS file also counts audio-only and video visits together, so it can't show which visit type people get. Person-level questions would need a survey such as HINTS, which we set aside for now.

> **Reproducibility.** The forecasting and audit code and tables are not in the files package. The results above are reproduced from the project report.

Limits, open questions and next steps

The main limit is that the AI evidence is thin and the state analysis can show association only.

- **Association, not cause.** State-level tests use about 47–50 states, so intervals are wide; they cannot say what happens to individuals.
- **Rural definitions differ.** CMS uses ZIP codes; the outside data uses counties. Connecticut lacks physician and Medicare Advantage data; New Jersey and Rhode Island have no rural county group.
- **Open question.** Why urban rates fell faster in states where many rural beneficiaries live is untested.
- **Next steps.** Test better features or a longer history for the learned models; review the Becker's-cited scribe study at its source; decide whether to add person-level survey data.

Data cleaning and checks

The CMS file has 36,120 rows (28,896 quarterly, 7,224 annual). Checks found no duplicates, no negative counts, no rates outside 0–100%, and every rate matches users divided by eligible users. 3,896 rows have a suppressed rate and 191 cells have fewer than 100 eligible users (kept and flagged). 31 quarterly rows show eligible users above Part B enrollment, flagged for review. Totals were reconciled to the CMS snapshot, charts were inspected after rendering, and results were re-run with a stricter rural definition. A person should still review before anything is shared.

Files in the package

The same material is in `telehealth_equity_project.zip`, with a README that gives the run order for the scripts.

FOLDER	FILES	WHAT IT HOLDS
report/	`AI_in_Telehealth_Who_Gets_Left_Out.docx`	Project report (Word export)
charts/	`fig1 – fig15 (PNG)`	All charts shown on this page
tables/	`national_quarterly_gap.csv`, `gap_trend_model.csv`	National rural and urban rates by quarter; gap trend model
tables/	`state_gap_2024.csv`, `stratified_gap_2024.csv`, `standardised_gap_2024.csv`	2024 state gaps, gaps within demographic groups, adjusted gaps
tables/	`state_quarterly_gaps.csv`, `state_gap_trends.csv`, `national_trend_robust.csv`	State gaps by quarter; per-state trends with Newey-West intervals
tables/	`gap_decomposition_by_year.csv`, `place_component_by_state_2024.csv`, `state_correlates_2024.csv`	Within-state vs place split of the gap; state contributions; state correlates
tables/	`state_merged_2024.csv`, `prespecified_associations.csv`, `multivariable_models.csv`, `national_rural_urban_factor_profile.csv`	Outside-data merge, hypothesis tests, models, rural vs urban county profile
tables/	`ratio_trend_by_weighting.csv`, `ratio_change_decomposition_by_state.csv`	Why state ratio trends and the national ratio disagree
tables/	`report_annual_benchmark.csv`	Annual totals from the CMS snapshot used to reconcile results
scripts/	`clean_telehealth.py`	Cleaning and validation of the CMS file
scripts/	`rural_urban_analysis.R`, `rural_urban_state_trends.R`, `rural_urban_drivers.R`	National analysis, state trends, gap decomposition
scripts/	`external_drivers.R, ratio_disagreement.R`	Outside-data tests; ratio disagreement analysis

FOLDER	FILES	WHAT IT HOLDS
data/	`telehealth_clean.csv, flagged_rows.csv, cleaning_log.md, raw CMS file, CMS snapshot PDF`	Cleaned data and its audit trail

Sources

CMS Medicare Telehealth Trends data file and 2026-09-14 snapshot report, supplied by the project owner.

ACS 2019-2023 5-year table-based summary file (tables B28005, B18101)

HRSA Area Health Resources Files 2024-2025

USDA Rural-Urban Continuum Codes 2023

Home-hospital voice assistant (Germans Trias)

Talk2Care (MIT)

AHRQ automated interpreting trial

Rural Health Information Hub: AI clinical documentation

Becker's: ambient AI and the Rural Health Transformation Program

Remote patient monitoring trial and CMS billing change

Speech recognition errors across accents (medRxiv preprint)

Digital health, AI and health equity review

Rural California telehealth modality study (JMIR 2025)

Medicare telehealth flexibilities extended through 2027 (Greenbaum)

Built from CMS Medicare Telehealth Trends data with analysis in R and Python. Claude assisted with the analysis and drafting.